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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PROCESS DATA SHEET: CENTRIFUGAL PUMP (API 610)

Document Code: DS-P2301 | Revision: Z1 | Status: Issued for Construction

Equipment Tag: P-2301 (P-2301A Operating / P-2301B Auto-Standby)

Equipment Name: Preflash Column Bottoms Transfer Pump

Equipment Class: Centrifugal Pump (API 610 Type OH2 Centerline Mounted)

Process Unit: Unit 2300 (Concentration & Decomposition Section)

Service: Transfers 82 wt% Concentrated CHP (Stream S229) from V-2301 Bottoms to Decomposer Reactor D-2304

1. HYDRAULIC & OPERATING DATA

- Pumped Fluid: Concentrated Cumene Hydroperoxide (82 wt% CHP in Cumene)
- Normal / Rated Volumetric Flow: 15.2 m³/hr / 18.5 m³/hr (Mass Flow 14,500 kg/hr)
- Pumping Temperature: 92.0 degC (Max Design Temperature: 165 degC)
- Specific Gravity at 92 degC: 0.955 | Viscosity: 1.45 cP
- Suction Pressure: 0.05 kg/cm²a (Vacuum Sump of V-2301)
- Discharge Pressure: 6.8 kg/cm²g
- Differential Head: 74.0 m | NPSHa: 4.2 m | NPSHr: 2.1 m

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DS-P2301 PAGE 2 OF 3 - MECHANICAL SEAL, METALLURGY & MOTOR DRIVER

Equipment Tag: P-2301

- API 610 Material Class: S-8 (All Wetted Parts 316L Stainless Steel / ASTM A351 Gr.CF3M)
- Mechanical Seal: API 682 Dual Pressurized Mechanical Seal, Plan 53B with clean Cumene barrier fluid
- Driver: 15.0 kW Explosion-Proof Induction Motor (Ex d IIB T3, 2950 RPM, 400V / 3Ph / 50Hz)

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Equipment Tag: P-2301

- Dead-Head Thermal Hazard Protection: Minimum Flow Orifice R0-0405 and Casing High Temperature Trip TSHH-0408 (S
- Tripped automatically by Interlock I-2304 on D-2304 High-High Temperature (TXSHH-1301 $\geq$ 115 degC).